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Chapter Review

Problems: Additional Problems

32. BIO Decades ago, it was thought that huge herbivorous dinosaurs such as *Apatosaurus* and *Brachiosaurus* habitually walked on the bottom of lakes, extending their long necks up to the surface to breathe. *Brachiosaurus* had its nostrils on the top of its head. In 1977, Knut Schmidt-Nielsen pointed out that breathing would be too much work for such a creature. For a simple model, consider a sample consisting of 10.0 L of air at absolute pressure 2.00 atm, with density $2.40\ \mathrm{kg/m^3}$, located at the surface of a freshwater lake. Find the work required to transport it to a depth of 10.3 m, with its temperature, volume, and pressure remaining constant. This energy investment is greater than the energy that can be obtained by metabolism of food with the oxygen in that quantity of air.

33. GP A helium-filled balloon (whose envelope has a mass of $m_b = 0.250\ \mathrm{kg}$) is tied to a uniform string of length $\ell = 2.00\ \mathrm{m}$ and mass $m = 0.050\ 0\ \mathrm{kg}$. The balloon is spherical with a radius of $r = 0.400\ \mathrm{m}$. When released in air of temperature $20°\mathrm{C}$ and density $\rho_{\text{air}} = 1.20\ \mathrm{kg/m^3}$, it lifts a length h of string and then remains stationary as shown in Figure P14.33. We wish to find the length of string lifted by the balloon.

(a) When the balloon remains stationary, what is the appropriate analysis model to describe it?

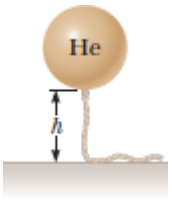
(b) Write a force equation for the balloon from this model in terms of the buoyant force B, the weight F_b of the balloon, the weight F_{He} of the helium, and the weight F_s of the segment of string of length h.

(c) Make an appropriate substitution for each of these forces and solve symbolically for the mass m_s of the segment of string of length h in terms of m_b, r, ρ_{air}, and the density of helium ρ_{He}.

(d) Find the numerical value of the mass m_s.

(e) Find the length h numerically.

Figure P14.33



34. S The true weight of an object can be measured in a vacuum, where buoyant forces are absent. A measurement in air, however, is disturbed by buoyant forces. An object of volume V is weighed in air on an equal-arm balance with the use of counterweights of density ρ. Representing the density of air as ρ_{air} and the balance reading as F_g', show that the true weight F_g is

$$F_g = F_g' + \left(V - \frac{F_g'}{\rho g}\right)\rho_{\text{air}}\, g$$

35. To an order of magnitude, how many helium-filled toy balloons would be required to lift you? Because helium is an irreplaceable resource, develop a theoretical answer rather than an experimental answer. In your solution, state what physical quantities you take as data and the values you measure or estimate for them.

36. QC **Review**. Assume a certain liquid, with density $1\,230\ \text{kg/m}^3$, exerts no friction force on spherical objects. A ball of mass 2.10 kg and radius 9.00 cm is dropped from rest into a deep tank of this liquid from a height of 3.30 m above the surface.

(a) Find the speed at which the ball enters the liquid.

(b) Evaluate the magnitudes of the two forces that are exerted on the ball as it moves through the liquid.

(c) Explain why the ball moves down only a limited distance into the liquid and calculate this distance.

(d) With what speed will the ball pop up out of the liquid?

(e) How does the time interval Δt_{down}, during which the ball moves from the surface down to its lowest point, compare with the time interval Δt_{up} for the return trip between the same two points?

(f) **What If?** Now modify the model to suppose the liquid exerts a small friction force on the ball, opposite in direction to its motion. In this case, how do the time intervals Δt_{down} and Δt_{up} compare? Explain your answer with a conceptual argument rather than a numerical calculation.

37. Evangelista Torricelli was the first person to realize that we live at the bottom of an ocean of air. He correctly surmised that the pressure of our atmosphere is attributable to the weight of the air. The density of air at $0°\text{C}$ at the Earth's surface is $1.29\ \text{kg/m}^3$. The density decreases with increasing altitude (as the atmosphere thins). On the other hand, if we assume the density is constant at $1.29\ \text{kg/m}^3$ up to some altitude h and is zero above that altitude, then h would represent the depth of the ocean of air.

(a) Use this model to determine the value of h that gives a pressure of 1.00 atm at the surface of the Earth.

(b) Would the peak of Mount Everest rise above the surface of such an

atmosphere?

38. BIO QIC A common parameter that can be used to predict turbulence in fluid flow is called the *Reynolds number*. The Reynolds number for fluid flow in a pipe is a dimensionless quantity defined as

$$\mathrm{Re} = \frac{\rho v d}{\eta}$$

where ρ is the density of the fluid, v is its speed, d is the inner diameter of the pipe, and η is the viscosity of the fluid. The criteria for the type of flow are as follows:

- If $\mathrm{Re} < 2\,300$, the flow is laminar.
- If $2\,300 < \mathrm{Re} < 4\,000$, the flow is in a transition region between laminar and turbulent.
- If $\mathrm{Re} > 4\,000$, the flow is turbulent.

(a) Let's model blood of density $1.06 \times 10^3\ \mathrm{kg/m^3}$ and viscosity $3.00 \times 10^{-3}\ \mathrm{Pa \cdot s}$ as a pure liquid, that is, ignore the fact that it contains red blood cells. Suppose it is flowing in a large artery of radius 1.50 cm with a speed of $0.067\,0\ \mathrm{m/s}$. Show that the flow is laminar.

(b) Imagine that the artery ends in a *single* capillary so that the radius of the artery reduces to a much smaller value. What is the radius of the capillary that would cause the flow to become turbulent?

(c) Actual capillaries have radii of about 5–10 micrometers, much smaller than the value in part (b). Why doesn't the flow in actual capillaries become turbulent?

39. In 1983, the United States began coining the one-cent piece out of copper-clad zinc rather than pure copper. The mass of the old copper penny is 3.083 g and that of the new cent is 2.517 g. The density of copper is $8.920\ \mathrm{g/cm^3}$ and that of zinc is $7.133\ \mathrm{g/cm^3}$. The new and old coins have the same volume. Calculate the percent of zinc (by volume) in the new cent.

40. S **Review**. With reference to the dam studied in Example 14.4 and shown in Figure 14.5,

(a) show that the total torque exerted by the water behind the dam about a horizontal axis through O is $\frac{1}{6}\rho g w H^3$.

(b) Show that the effective line of action of the total force exerted by the water is at a distance $\frac{1}{3}H$ above O.

41. The *spirit-in-glass thermometer*, invented in Florence, Italy, around 1654, consists of a tube of liquid (the spirit) containing a number of submerged glass spheres with slightly different masses (Fig. P14.41). At sufficiently low

temperatures, all the spheres float, but as the temperature rises, the spheres sink one after another. The device is a crude but interesting tool for measuring temperature. Suppose the tube is filled with ethyl alcohol, whose density is $0.789\ 45\ \mathrm{g/cm^3}$ at $20.0°\mathrm{C}$ and decreases to $0.780\ 97\ \mathrm{g/cm^3}$ at $30.0°\mathrm{C}$.

(a) Assuming that one of the spheres has a radius of 1.000 cm and is in equilibrium halfway up the tube at $20.0°\mathrm{C}$, determine its mass.

(b) When the temperature increases to $30.0°\mathrm{C}$, what mass must a second sphere of the same radius have to be in equilibrium at the halfway point?

(c) At $30.0°\mathrm{C}$, the first sphere has fallen to the bottom of the tube. What upward force does the bottom of the tube exert on this sphere?

Figure P14.41

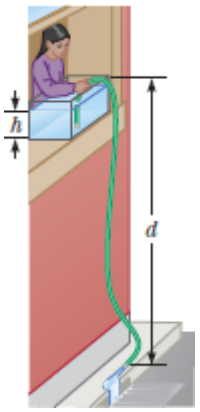


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42. **S** A woman is draining her fish tank by siphoning the water into an outdoor drain as shown in Figure P14.42. The rectangular tank has footprint area A and depth h. The drain is located a distance d below the surface of the water in the tank, where $d \gg h$. The cross-sectional area of the siphon tube is A'. Model the water as flowing without friction. Show that the time interval required to empty the tank is given by

$$\Delta t = \frac{Ah}{A'\sqrt{2gd}}$$

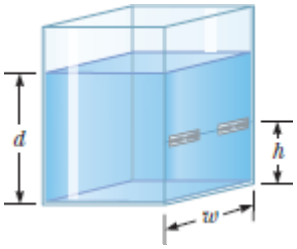
Figure P14.42



43. CR NS **Review**. You and your father are designing a waterfall for your backyard swimming pool. At the top of the waterfall is a tank containing water that is kept to a depth $d = 0.280\ \text{m}$ by a pump. As shown in Figure P14.43, there is a small hatch of height $h = 0.100\ \text{m}$ and width $w = 0.150\ \text{m}$, hinged at the top, that can be used to turn on and turn off the supply of water to the waterfall. You want to attach a simple latch at the center of the bottom of the hatch and are trying to decide what type of latch to buy at your local hardware store. To make that decision, you need to determine the force that the latch must be able to withstand to keep the hatch closed.

Figure P14.43

Problems 43 and 44.



44. NS **Review**. You and your father are designing a waterfall for your backyard swimming pool. At the top of the waterfall is a tank containing water that is kept to a depth d by a pump. As shown in Figure P14.43, there is a small hatch of height h and width w, hinged at the top, that can be used to turn on and turn off the supply of water to the waterfall. You want to attach a simple latch at the center of the bottom of the hatch and are trying to decide what type of latch to buy at your local hardware store. To make that decision, you need to determine the force that the latch must be able to withstand to keep the hatch closed.

45. **Review**. A uniform disk of mass 10.0 kg and radius 0.250 m spins at 300 rev/min on a low-friction axle. It must be brought to a stop in 1.00 min by a brake pad that makes contact with the disk at an average distance 0.220 m from the axis. The coefficient of friction between pad and disk is 0.500. A piston in a cylinder of diameter 5.00 cm presses the brake pad against the disk. Find the pressure required for the brake fluid in the cylinder.

46. QC **Review**. In a water pistol, a piston drives water through a large tube of area A_1 into a smaller tube of area A_2 as shown in Figure P14.46. The radius of the large tube is 1.00 cm and that of the small tube is 1.00 mm. The smaller tube is 3.00 cm above the larger tube.

(a) If the pistol is fired horizontally at a height of 1.50 m, determine the time interval required for the water to travel from the nozzle to the ground. Neglect air resistance and assume atmospheric pressure is 1.00 atm.

(b) If the desired range of the stream is 8.00 m, with what speed v_2 must the stream leave the nozzle?

(c) At what speed v_1 must the plunger be moved to achieve the desired

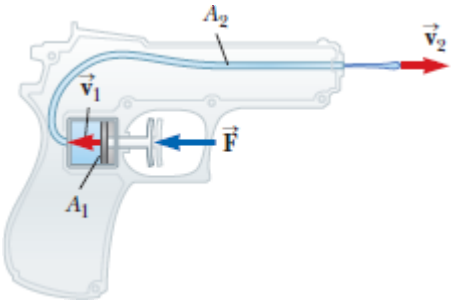
range?

(d) What is the pressure at the nozzle?

(e) Find the pressure needed in the larger tube.

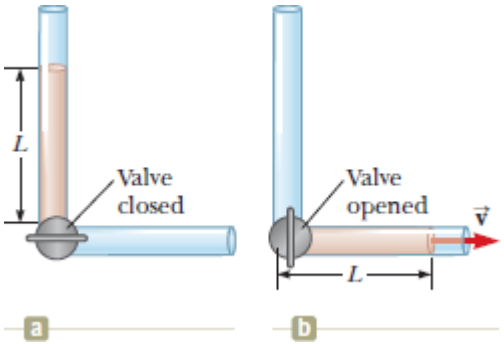
(f) Calculate the force that must be exerted on the trigger to achieve the desired range. (The force that must be exerted is due to pressure over and above atmospheric pressure.)

Figure P14.46



47. An incompressible, nonviscous fluid is initially at rest in the vertical portion of the pipe shown in Figure P14.47a, where $L = 2.00\ \text{m}$. When the valve is opened, the fluid flows into the horizontal section of the pipe. What is the fluid's speed when all the fluid is in the horizontal section as shown in Figure P14.47b? Assume the cross-sectional area of the entire pipe is constant.

Figure P14.47



48. S The hull of an experimental boat is to be lifted above the water by a hydrofoil mounted below its keel as shown in Figure P14.48. The hydrofoil has a shape like that of an airplane wing. Its area projected onto a horizontal surface is A. When the boat is towed at sufficiently high speed, water of density ρ moves in streamline flow so that its average speed at the top of the hydrofoil is n times larger than its speed v_b below the hydrofoil.

(a) Ignoring the buoyant force, show that the upward lift force exerted by the water on the hydrofoil has a magnitude

$$F \approx \frac{1}{2}(n^2 - 1)\rho v_b^2 A$$

(b) The boat has mass M. Show that the liftoff speed is given by

$$v \approx \sqrt{\frac{2Mg}{(n^2 - 1)A\rho}}$$

Figure P14.48



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